Timing Search[†]

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Abstract

Why do different industries fluctuate at different frequencies? Because each industry has its own profit curvature (φ) and timing friction (μ), and these two primitives—not the shock process—set the cycle length. I develop a *timing-search* model: agents choose when to enter a market under frictions that penalize rapid adjustment. The indifference condition across entry dates prevents the industry from settling at its optimum—entry must persist past optimal scale, forcing a return—and cycles arise at frequency $\omega = \sqrt{\varphi/\mu}$. Cycle frequency and the per-firm welfare cost of cycles have disjoint primitives: market structure (φ, μ) sets frequency; shock variance and bottleneck dissipation set welfare. Empirically, the predicted slope is recovered across 16 manufacturing industries with historical concentration as instrument; entry inputs flow flexibly across industries while the profit landscape does not, suggesting the profit landscape as the operative cross-industry channel. Entry-exit oscillations, post-shock periodicity, and a post-COVID overshoot are additional consistent diagnostics.

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1 Introduction

U.S. industries fluctuate at sharply different frequencies: dominant spectral periods range from 2.6 to 6.6 years in industrial production and from 2.4 to nearly 10 years in establishment entry. Why do different industries fluctuate at different frequencies? Standard business-cycle frameworks derive cycle properties from a common shock process (Kydland and Prescott, 1982a) or policy rule (Christiano et al., 2005), leaving cross-industry frequency variation unexplained. Two obstacles stand in the way of a structural account. First, in models where cycle frequency is inherited from shock persistence or policy, industries hit by common shocks should share a common cycle length—a prediction the data reject. Second, existing endogenous-cycle models generate internal persistence but do not deliver an industry-decomposable frequency formula or a separation between the determinants of timing and welfare. This paper overcomes both obstacles by introducing *timing search*—agents choosing *when* to enter a market under frictions that prevent instantaneous adjustment—and showing that the equilibrium cycle frequency takes the closed form

$$\omega \;=\; \sqrt{\varphi/\mu}, \tag{1}$$

in two market-structure primitives, while the per-firm welfare cost depends on a disjoint pair $(\sigma_\varepsilon, \chi)$. The indifference requirement that pins ω rules out a quiet return to optimal scale, generating cycles whose pitch is fixed by market structure but whose stationary amplitude is selected through a separate Rayleigh-friction channel.

This prediction should not be read as a claim that industries move in regular calendar-time cycles. Real industries are hit by many shocks, and observed cycles are noisy and intermittent. The claim is instead that the adjustment system has a characteristic frequency: after sufficiently large disturbances, the response should contain a frequency component whose location is governed by market-structure primitives.

The analysis delivers three results. First, I develop a mechanism for industry-specific cycles: timing search turns the entry-date choice into a source of endogenous oscillation. The central equilibrium restriction is an indifference condition across entry dates—a temporal analog of the equal-profit condition in Burdett and Mortensen (1998)—which equalizes entry value across dates and forces the industry to overshoot optimal scale rather than settle monotonically. Under Gaussian profits and an exponential-quadratic timing friction, the equilibrium reduces to a conservative

harmonic oscillator with period exactly $2\pi\sqrt{\mu/\varphi}$ within the all-entry region (Theorem 1, Section 3), where φ is the curvature of the profit landscape and μ is the timing friction from congestion in entry matching. The formula extends locally to general hump-shaped profits (Proposition 3).

Second, I prove a frequency-welfare separation that decomposes what determines cycle timing from what determines cycle welfare cost (Theorem 2, Section 4). Convex congestion in the entry-replacement bottleneck—a Rayleigh friction with coefficient χ structurally distinct from the per-firm sunk cost κ—turns the conservative orbit into a damped stochastic oscillator whose stationary amplitude is selected from primitives. The structural frequency $\omega_0 = \sqrt{\varphi/\mu}$ depends only on (φ, μ), while the per-firm effective-output welfare cost $\mathbb{E}[\mathcal{L}] = \sigma_\varepsilon^2/(2\chi)$ depends only on the disjoint pair $(\sigma_\varepsilon, \chi)$. The decomposition delivers a sharp policy anatomy: changing the speed of cycles is not the same as changing their welfare cost. Policies that move μ affect timing alone; policies that move φ affect timing and cycle severity but not expected per-firm welfare under the maintained metric; only policies that move χ or σ_ε affect welfare (Section 6.2, Section 6.3). The separation is exact under linear utility and the per-firm metric, with numerical leakage of order 10–25% under CRRA preferences or welfare metrics that include the supplier's resource flow (Online Appendix E.1).

Third, I bring the prediction that cycle frequency is pinned by market structure rather than by shocks to U.S. industry data (Section 5). The headline test regresses the spectral peak frequency on the Ellison and Glaeser (1997) geographic concentration index γ_i across 16 manufacturing 3-digit industries. The OLS slope is $\hat{\beta}_\gamma = 0.20$ ($p = 0.010$, $R^2 = 0.39$); instrumenting current γ_i with historical Hoover localization coefficients from Kim (1995) raises the estimate to $\hat{\beta}_\gamma^{\mathrm{IV}} = 0.45$, statistically indistinguishable from the structural prediction $\beta_\gamma = 1/2$ under both Wald and Anderson-Rubin weak-IV-robust inference. The first stage is weak ($F = 1.55$), so I read the IV as corroborative rather than independently identifying. The same sign holds in a parallel BDS subsector cross-section using independent data on both cycle frequency and concentration.

This evidence is consistent with—but does not directly identify—a φ-driven cross-industry channel. External proxies for the matching friction (capital intensity, asset durability, regulatory burden) deliver wrong-signed coefficients in the cross-industry regression (Online Appendix D.5); this would arise either under approximate μ-uniformity within manufacturing or if those proxies fail to capture the friction-relevant

component of μ. The recovered slope's broad consistency with the structural $\beta_\gamma = 1/2$ corroborates the same reading but is not independently identifying given the weak first stage. The maintained identifying restriction—that entry inputs flow flexibly across 3-digit manufacturing industries while the profit landscape firms enter does not—has suggestive support from these patterns; a direct test of the μ channel with industry-friction-specific proxies remains for future work.

Three additional diagnostics are consistent with the mechanism but rest on weaker identification. BDS net-entry rates show oscillatory features outside the AR(1) class in 10 of 19 sectors. Manufacturing and oil investment oscillate at industry-specific frequencies after large historical shocks ($T = 5.0$ years post-WWII; $T = 3.25$ years post-1979). And 14 of 19 BDS sectors overshoot above their pre-pandemic trend after COVID-19. Against this evidence, a side-by-side simulation of AR(1) RBC, New Keynesian AR(2), and Beaudry et al. (2020) benchmarks, together with a multi-sector RBC envelope with input-output linkages and AR(2) productivity shocks, shows that none of these alternatives jointly produces cross-industry frequency heterogeneity, post-shock periodicity, and overshoot at the magnitudes observed (Online Appendix F.1, F.2).

Related literature This paper builds on five strands. On endogenous business cycles, Beaudry et al. (2020) generate a stochastic limit cycle from financial frictions via Hopf bifurcation, with a recent extension in Beaudry et al. (2024); Beaudry and Portier (2006) document stock prices leading TFP; Grandmont (1985), Azariadis (1981), Benhabib and Farmer (1994), and Goodwin (1967) provide earlier endogenous-cycle precedents. The closest precedent for the present mechanism is Jovanovic (2009), who derives endogenous business cycles from the option value of investment timing; the present paper extends the timing-driven logic to the entry-date choice and obtains a closed-form frequency in market-structure primitives. On search theory, timing search extends Stigler (1961), McCall (1970), and especially Burdett and Mortensen (1998) to the time domain: agents choose among dates rather than among contemporaneous alternatives, and the ironing condition is the temporal analog of the equal-profit condition in Burdett and Mortensen (1998) (Online Appendix A.1). On industry dynamics, the model adds timing frictions to Hopenhayn (1992) and the policy-extended Hopenhayn and Rogerson (1993) framework, and complements Jovanovic (1982)'s learning-based exit explanation by generating endogenous entry waves through

the ironing condition; Davis and Haltiwanger (1992) document the entry/exit empirics this literature studies. On agglomeration, the scale-congestion logic of Krugman (1991), Duranton and Puga (2004), Rosenthal and Strange (2004), Ellison and Glaeser (1997), and Ellison et al. (2010) is transposed from space to time: combined with timing frictions, it generates temporal patterns (cycles) rather than spatial patterns (clusters), and the EG concentration index provides the empirical proxy for φ; Syverson (2004) provides direct evidence of the productivity-density relationship that underlies the hump-shaped profit assumption. Finally, the micro-foundation $\mu = (1-\alpha)/[\alpha(\delta n_p)^2]$ connects to the Diamond-Mortensen-Pissarides literature (Mortensen and Pissarides, 1994; Diamond, 1982; Pissarides, 2000; Petrongolo and Pissarides, 2001; Shimer, 2005): matching frictions in entry generate equilibrium business cycles in the same way that DMP matching frictions generate equilibrium unemployment. A separate strand on aggregate fluctuations from disaggregated origins (Gabaix, 2011; Acemoglu et al., 2012) provides the natural alternative the paper tests against in Online Appendix F.2.

2 Environment

The environment has three elements chosen to make the cycle-frequency result precise: a hump-shaped profit landscape that gives the industry a well-defined optimal scale, a matching function for entry that endogenously generates a quadratic adjustment cost, and linear-utility households that let industries evolve independently. Section 2.1 sets up the agents, Section 2.2 the profit landscape, Section 2.3 the timing friction, and Section 2.4 combines them into the reduced-form dynamics that drive the equilibrium.

2.1 Agents

The economy consists of two types of agents: a representative household that consumes and supplies labor, and a fixed mass of potential entrepreneurs who alternate between operating firms and searching for entry opportunities. The household has linear utility over consumption, $U = \int_0^\infty e^{-\rho t} C(t)\,\mathrm{d}t$, and supplies labor perfectly elastically at a fixed wage $\bar{w}$. Linear utility is not merely a simplification: it eliminates wealth effects and cross-industry price feedback, making the single-industry analysis a general equilibrium result (see below). Three ingredients define the entry side of the environment.

Assumption 1 (Entry economy)**.**
A mass $M > 0$ of potential entrepreneurs, each at any instant either active *(operating a firm) or* searching *(a potential entrant). Active firms produce using labor hired from the representative household at wage $\bar{w}$, with technology $y_i = zl_i^{\gamma}$, $\gamma \in (0, 1)$. Firms exit at Poisson rate $\delta > 0$, returning to the search pool.*

General equilibrium Because the wage is fixed and households have linear utility, each industry's profit landscape depends only on its own number of active firms. Under idiosyncratic shocks and no inter-industry input-output linkages or demand spillovers, the multi-industry economy with I industries each characterized by (φ_i, μ_i) has industries that evolve independently, with aggregate spectral density a superposition of industry-specific peaks at frequencies $\omega_i = \sqrt{\varphi_i/\mu_i}$. The isolation of frequency from the shock process is therefore a *cross-sectional* prediction: different industries cycle at different frequencies determined by their respective market structures. Under common shocks or input-output linkages, the spectral superposition does not survive aggregation. Section 6 bounds the scope of the closed-form result, with quantitative leakage of (φ, μ) into the welfare cost under CRRA preferences documented in Online Appendix E.1. Exogenous exit at rate δ ensures a steady flow of agents returning to the search pool, maintaining the timing search problem at every date.

Let $n(t)$ denote the mass of active production units at date t and $S(t) = M - n(t)$ the mass of searchers. Throughout the paper, I interpret n as the mass of active *establishments*—physical production units such as plants, stores, or branches—rather than firms (legal entities). The establishment interpretation matches both empirical series used in Sections 5–5.4: BEA investment in structures measures the rate of new establishment construction, and BDS establishment entry and exit rates measure establishment births and deaths. The matching-congestion microfoundation of μ (Section 2.3) is also most natural at the establishment level, where bottlenecks in permits, specialist labor, equipment, and sites constrain rapid adjustment. The aggregate state of the economy is fully characterized by $n(t)$ and its rate of change $\dot{n}(t)$.

2.2 Matching efficiency and the profit landscape

Each firm's effective output depends on market thickness. More firms create denser networks of suppliers, customers, and complementary services, improving efficiency.

But beyond a point, additional firms congest the market. A scale efficiency function $m(n)$ multiplies each firm's output: $y_i = zl_i^\gamma \cdot m(n)$. Profit is $\pi(n) = \bar{\pi}(w) \cdot m(n)$, where $\bar{\pi}$ is the standard profit given the wage.

Assumption 2 (Gaussian scale efficiency)**.**
Scale efficiency takes the Gaussian form:

$$m(n) = \exp\Big(-\frac{\varphi}{2}(n - n_p)^2\Big), \qquad \varphi > 0,\ n_p > 0. \tag{2}$$

Interpretation of Assumption 2. The function peaks at $n = n_p$ (optimal market scale) and falls off symmetrically. The parameter φ governs the sharpness of the trade-off: high φ means a narrow peak (sharp trade-off); low φ means a broad peak (robust to deviations). Online Appendix A.2 provides a matching-theoretic microfoundation.

For the analytical results, I work with the reduced-form Gaussian profit:

$$\pi(n) = \pi_0 \exp\Big(-\frac{\varphi}{2}(n - n_p)^2\Big), \qquad \pi_0 > 0. \tag{3}$$

The semi-elasticity is exactly linear:

$$f(n) \equiv \frac{\pi'(n)}{\pi(n)} = -\varphi(n - n_p). \tag{4}$$

This linearity delivers exact harmonic oscillations (Theorem 1). The general case—where wage congestion adds a non-quadratic correction to $\ln \pi$—produces anharmonic oscillations with the same leading-order frequency (Proposition 3).

2.3 The timing friction: congestion in the entry process

Entry is mediated by a matching function. The $S(t) = M - n(t)$ searchers compete for R units of *entry resources*—permits, specialized labor, venture capital, factory sites—to form matches:

Assumption 3 (Entry matching)**.** $m(S, R) = S^\alpha R^{1-\alpha}$, $\alpha \in (0, 1)$.

The matching infrastructure is tuned to clear the steady-state throughput $e_{ss} = \delta n_p$. Following the convention introduced in Section 2.5, baseline replacement is serviced smoothly by a layer of capacity that scales with the prevailing population, while the

timing friction applies to net adjustment pressure $\dot{n}$—deviations of the adjustment-speed margin from zero. Rapid entry ($\dot{n} > 0$) overloads permitting and construction; rapid net contraction ($\dot{n} < 0$) overloads resale and reorganization. Both reduce the matching probability per searcher and produce the timing friction.

Proposition 1 (Micro-foundation of the timing friction)**.**
Under Assumptions 1–3, the entry matching friction produces, to second order around the steady state, the timing friction $\xi(\dot{n}) = e^{-\frac{\mu}{2}\dot{n}^2}$ *with*

$$\mu = \frac{1-\alpha}{\alpha\,(\delta n_p)^2}. \tag{5}$$

Sketch. A symmetric bottleneck-strain penalty $g(\dot{n}^2/(\delta n_p)^2)$ with $g(0) = 0$ and $g'(0) = 1/2$ multiplies the Cobb-Douglas matching probability; we adopt the specific functional form $g(z) = \ln(1+z/2)$ (yielding $\xi(\dot{n}) = [1+\dot{n}^2/(2(\delta n_p)^2)]^{-(1-\alpha)/\alpha}$) as a representative element of this equivalence class. Taylor expansion at second order gives $\ln \xi(\dot{n}) = -\frac{\mu}{2}\dot{n}^2 + O(\dot{n}^4)$ with $\mu = (1-\alpha)/[\alpha(\delta n_p)^2] > 0$. *The symmetry of g in $\dot{n}$ is an additional knife-edge condition* beyond Assumption 3: under asymmetric upside (permitting/construction) and downside (resale/reorganization) strain, with second-order curvatures $\mu_+ \neq \mu_-$, the leading-order ironing condition would carry a cubic term in $\dot{n}$ that produces amplitude-dependent frequency, of the same general form analyzed in Proposition 5. The exact-harmonic result of Theorem 1 requires the symmetric specification; the qualitative cycle survives asymmetric strain. Online Appendix A.4 provides the detailed derivation. ■

The timing friction μ decreases in the matching elasticity α and decreases in the steady-state throughput δn_p (the entry-resource stock R is absorbed into the steady-state calibration of n_p via the matching technology and therefore does not appear in (5) as a separate factor). Substituting (5) into $\omega = \sqrt{\varphi/\mu}$ gives the structural frequency

$$\omega = \delta n_p \sqrt{\alpha\varphi/(1-\alpha)}, \tag{6}$$

connecting cycle frequency to three independently measurable objects: matching elasticity α (estimated in the range $[0.5, 0.7]$ for labor markets, Petrongolo and Pissarides 2001), profit curvature φ, and steady-state turnover δn_p. Effective output is $y(t) = \pi(n(t))\,\xi(\dot{n}(t))$. The quadratic form in $\dot{n}$ arises generically from any smooth

symmetric matching friction; Online Appendix A.3 derives the same ξ from discrete-time proportional adjustment costs.

2.4 Combined structure

Combining the Gaussian profit and the Gaussian friction, effective output is:

$$y(t) = \pi_0 \exp\Big(-\frac{\varphi}{2}(n - n_p)^2 - \frac{\mu}{2}\dot{n}^2\Big) . \tag{7}$$

The effective output is a bivariate Gaussian kernel in $(n, \dot{n})$, centered at $(n_p, 0)$. Both terms impose quadratic penalization of deviation from the optimum: n_p for market scale and $\dot{n} = 0$ for adjustment speed.

2.5 Infrastructure supplier and the Rayleigh friction χ

The matching technology of Section 2.3 treats the entry-resource stock as fixed. In practice it is supplied by a separate sector—permitting authorities, construction labor, specialist equipment manufacturers—that produces resources at a flow cost convex in the rate of provision. We adopt a two-layer convention: a baseline layer that scales smoothly with $n(t)$ to service replacement $\delta n(t)$, and a bottleneck layer that services the net adjustment pressure $\dot{n}(t)$ at quadratic congestion cost $\frac{1}{2}c_2(R - R_{ss})^2$ with linear linkage $R(t) - R_{ss} = r\,\dot{n}(t)$. Substituting yields a quadratic resource-cost deviation $\frac{1}{2}c_2 r^2\,\dot{n}^2$. Defining the Rayleigh friction coefficient

$$\chi \equiv c_2\, r^2, \tag{8}$$

this flow equals $\frac{1}{2}\chi\,\dot{n}^2$ and contributes a damping force $-\chi\dot{n}$ to the equation of motion via the Lagrangian dissipation function $\mathcal{R}(\dot{n}) = \frac{1}{2}\chi\dot{n}^2$, selecting the stationary cycle amplitude in Proposition 7. The three coefficients (κ, μ, χ) map to three different primitives: κ is the per-firm sunk entry cost, paid by entrants and pinning the steady-state ironing level; μ is the matching friction inside the multiplicative kernel $\xi(\dot{n})$, pinning the cycle frequency $\omega = \sqrt{\varphi/\mu}$; χ is the supplier-side Rayleigh coefficient, pinning the cycle amplitude. The $\frac{1}{2}\chi\dot{n}^2$ flow is a real resource cost paid by households for infrastructure services, separate from the per-firm effective-output welfare metric of Section 6.2; it enters the welfare formula only through its damping effect on equilibrium amplitude. Online Appendix B.5 provides the full setup, the supplier's first-order

condition, the alternative single-bottleneck specification, and the energy-balance derivation.

3 Ironing Along Time: Equilibrium and the Main Result

The equilibrium condition that agents cannot profit by retiming their entry—the ironing condition—pins a rate of entry along any approach to optimal scale. That pinned rate becomes momentum the industry cannot discard at the optimum without creating arbitrage. The result is cycles, and in the frictionless limit they are exactly harmonic with frequency $\omega = \sqrt{\varphi/\mu}$. This section formalizes the argument.

3.1 The agent's problem

A searching agent chooses a date τ at which to enter. Upon entry, she operates a firm earning effective flow payoff $\pi(n(t))\xi(\dot{n}(t))$ per unit time, discounted at rate $r \geq 0$, with exogenous exit at rate $\delta > 0$. The value of entering at date τ is:

$$V(\tau) = \int_{\tau}^{\infty} e^{-(r+\delta)(t-\tau)}\, \pi(n(t))\, \xi(\dot{n}(t))\, \mathrm{d}t. \tag{9}$$

The agent maximizes $V(\tau)$ over τ, taking the aggregate path $\{n(t)\}$ as given. An agent who does not enter at date t remains in the search pool, with continuation value

$$V^{\text{search}}(t) = \max_{\tau \geq t} e^{-r(\tau-t)}\big[V(\tau) - \kappa\big], \tag{10}$$

where $\kappa \geq 0$ is the per-firm sunk entry cost paid at the moment of entry. The entry-matching infrastructure (permits, construction crews, specialist services) is also bottlenecked, with strain rising in the squared departure of net adjustment pressure $\dot{n}$ from zero; this strain is summarized at the macroscopic level by a Rayleigh friction coefficient $\chi > 0$, which produces a damping force $-\chi\dot{n}$ in the equilibrium equation of motion (Online Appendix B.5). Two parameters are therefore distinct in the model: κ is the dollar amount the marginal entrant pays at entry; χ is the macroscopic damping coefficient governing the bottleneck-strain mechanism. The constant and linear parts of the entry cost (including κ) pin the ironing level but do not dissipate;

the Rayleigh friction χ generates the damping $\eta = \chi/\mu$ that appears in the operative equilibrium. We treat χ as a reduced-form amplitude-selection primitive: it enters the equilibrium through the damping force in the equation of motion, and through the welfare identity (25) only via the equilibrium amplitude. The supplier's resource flow $\frac{1}{2}\chi\dot{n}^2$ is a real-but-separate accounting object documented in Section 2.5; it is not part of the per-firm effective-output welfare metric. We track κ and χ separately throughout: κ enters the free-entry condition above, while χ enters the damped equation of motion and the welfare formula.

3.2 Equilibrium

An *entry policy* is a measurable function $e : \mathbb{R}_+ \to \mathbb{R}_+$ specifying the flow rate of entrants at each date. Together with the initial condition $n(0)$, it determines the entire path of the economy through the aggregate consistency condition $\dot{n} = e - \delta n$.

Definition 1 (Timing search equilibrium)**.**
A timing search equilibrium *is a path* $\{n(t)\}_{t\geq 0}$ *and an entry policy* $e(\cdot)$ *such that:*

(i) ***Individual optimality****: Each agent chooses her entry date to maximize* $V(\tau)$*.*

(ii) ***Temporal free entry (with complementary slackness)****:* $V(t) \leq V^{search}(t) + \kappa$ *at all dates, with equality whenever* $e(t) > 0$ *and* $n(t) < M$*. At dates where the population cap binds* ($n(t) = M$) *or the entry flow is zero* ($e(t) = 0$)*, the inequality may be strict.*

(iii) ***No profitable timing deviation****:* $V(\tau) \leq V(t)$ *for all* τ *and all* t *in the support of entry.*

(iv) ***Aggregate consistency****:* $\dot{n}(t) = e(t) - \delta\, n(t)$*, where* $e(t) \geq 0$*.*

(v) ***Feasibility****:* $n(t) \leq M$ *for all* t*.*

The no-profitable-timing-deviation condition Condition (iii) is the key requirement. In Burdett and Mortensen (1998), the analogous condition is that no firm can profit by reposting its wage, which yields equal profits across wages. Here, the condition that no agent can profit by retiming her entry yields equal profits across dates. If any date offered strictly higher value, agents would crowd that date, changing $n(t)$ until the advantage was eliminated.

Commitment versus dynamic optimization The formulation in (9)–(10) is a commitment problem: agents choose τ taking the entire future path $\{n(t)\}$ as given. An alternative formulation would have agents observe the current state $(n(t), \dot{n}(t))$ at each instant and decide dynamically whether to enter. The ironing condition makes the two equivalent. Since $V(\tau) = V^*$ for all τ in equilibrium, an agent who could re-optimize at any date t would find no profitable deviation—she is indifferent across all entry dates. The commitment equilibrium is therefore also an equilibrium of the dynamic problem: agents who can freely re-time their entry choose not to.

3.3 The ironing condition

The key equilibrium condition is that agents cannot profit by retiming their entry. This is the temporal analog of the equal-profit condition in Burdett and Mortensen (1998): Burdett-Mortensen equalize profits across wages so that no firm gains by reposting, and timing search equalizes payoffs across dates so that no agent gains by retiming. I call this the *ironing condition*, since it "irons out" any temporal variation in the value of entry.

Proposition 2 (Ironing condition)**.**
In a timing search equilibrium where entry occurs at all dates, agents are indifferent across entry dates. Equivalently, the effective flow payoff is equalized:

$$\pi(n(t))\,\xi(\dot{n}(t)) = C \qquad \textit{for all } t, \tag{11}$$

for some constant $C > 0$.

Proof. Suppose entry occurs at all dates. For any two dates τ_1, τ_2 in the support of entry, condition (iii) of Definition 1 requires $V(\tau_1) \leq V(\tau_2)$ and $V(\tau_2) \leq V(\tau_1)$, so $V(\tau_1) = V(\tau_2)$. Since the support is all of $\mathbb{R}_+$, $V(\tau) = V^*$ for all $\tau \geq 0$. The equilibrium path $n(\cdot)$ is C^1 by aggregate consistency (condition (iv)) and the smoothness of π and ξ, so $V(\tau)$ is differentiable in τ.[1] Differentiating (9):

$$\frac{\mathrm{d}V}{\mathrm{d}\tau} = -\pi(n(\tau))\,\xi(\dot{n}(\tau)) + (r+\delta)\,V(\tau).$$

[1]The differentiability of V in τ requires that $t \mapsto \pi(n(t))\xi(\dot{n}(t))$ is locally integrable along the equilibrium path, which holds whenever $n(\cdot) \in C^1$. This regularity is guaranteed by the smoothness of π (Assumption 2) and ξ (Section 2.3), combined with the continuity of the entry flow $e(\cdot)$ in condition (iv).

Setting $\frac{\mathrm{d}V}{\mathrm{d}\tau} = 0$ (since V is constant): $\pi(n(\tau))\xi(\dot{n}(\tau)) = (r+\delta)V^* \equiv C$. ■

Regularity condition The condition that entry occurs at all dates is satisfied whenever the searcher pool $S(t) = M - n(t)$ is bounded away from zero along the equilibrium path and the timing friction μ is positive. With a large population M and moderate cycle amplitude $A < M - n_p$, the condition holds. Online Appendix B.3 provides a formal verification.

3.4 The main result

The equilibrium under Gaussian profits and no dissipation is an exact harmonic oscillator with frequency $\omega = \sqrt{\varphi/\mu}$ (Theorem 1). This subsection derives that result. The frictionless case is not the most realistic specification, but it is the analytically tractable one: the ironing condition reduces to a conservative harmonic oscillator whose orbit and period admit closed-form expressions at every amplitude.

The closed-form frequency carries over to the full operative equilibrium with Rayleigh friction and shocks, which is a damped, stochastically forced version of the same oscillator:

$$\ddot{n} + \eta\,\dot{n} + \frac{\varphi}{\mu}\,(n - n_p) = \frac{\sigma_\varepsilon}{\mu}\,\varepsilon_t, \tag{12}$$

with $\eta = \chi/\mu$ the damping rate from convex congestion in entry processing (with χ the Rayleigh friction coefficient introduced above) and σ_ε the shock volatility. Section 4 characterizes this damped-stochastic equilibrium and the stationary amplitude it selects. All three cases—frictionless, damped without shocks, damped with shocks—share the same natural frequency ω and differ only in whether the oscillator is isolated, decaying, or continuously excited.

Taking logarithms of the ironing condition (11) and substituting the functional forms (7):

$$-\frac{\varphi}{2}(n - n_p)^2 - \frac{\mu}{2}\,\dot{n}^2 = E, \tag{13}$$

where $E = \ln C - \ln \pi_0 < 0$ is constant along the equilibrium path. Differentiating with respect to t and dividing by $\dot{n} \neq 0$:

$$\ddot{n} = -\frac{\varphi}{\mu}\,(n - n_p). \tag{14}$$

Theorem 1 (Exact harmonic oscillations)**.**
In a timing search equilibrium (Definition 1) under Assumptions 1–3 together with the symmetric exponential-quadratic timing friction of Proposition 1 ($\xi(\dot{n}) = e^{-\mu\dot{n}^2/2}$ with $\mu = (1-\alpha)/[\alpha(\delta n_p)^2]$), and provided that entry occurs at all dates (the regularity condition formalized in the verification paragraph below, equivalent to $A \leq \delta n_p/\sqrt{\omega^2+\delta^2}$), the equilibrium path is:

$$n(t) = n_p + A\cos(\omega t + \phi), \qquad \omega = \sqrt{\frac{\varphi}{\mu}}. \tag{15}$$

The solution is exact within the feasible all-entry region and requires no linearization. The amplitude A and phase ϕ are determined by initial conditions (subject to the feasibility bound), and the ironing condition holds with $E = -\frac{\varphi}{2}A^2$.

Proof. The ironing condition (Proposition 2) gives (13). Differentiation yields the harmonic oscillator equation (14), whose general solution is (15) with $\omega = \sqrt{\varphi/\mu}$. The solution is exact because the Gaussian specification makes the semi-elasticity $f(n)$ exactly linear (equation (4)), so no Taylor approximation is needed. Substituting back: $\frac{\varphi}{2}A^2\cos^2(\omega t+\phi) + \frac{\mu}{2}\omega^2 A^2 \sin^2(\omega t+\phi) = \frac{\varphi}{2}A^2 = -E$. ■

Verification, scope, and selection The construction establishes existence: for any amplitude $A \leq \delta n_p/\sqrt{\omega^2+\delta^2}$ (the binding sufficient condition for $e(t) \geq 0$ throughout the cycle, equivalent to $\delta(n_p - A) \geq \omega A$), the pair $(n(t), e(t))$ is a timing search equilibrium, with the entry policy $e(t) = \dot{n}(t) + \delta n(t)$ non-negative and feasibility holding for $M > n_p + A$ (Online Appendix B.6). The condition is tighter than the looser $S(t) > 0$ requirement that searchers be available, and it sharpens the empirical-applicability scope: industries whose net establishment entry rate routinely goes substantially negative (i.e., $|\dot{n}| > \delta n_p/2$) are outside the regime where the ironing condition holds globally, and the equilibrium of Proposition 2 should be read as applying to its complement. The center structure leaves the amplitude A free; the stochastic extension (Proposition 6) and the dissipation result (Proposition 7) resolve this multiplicity by selecting a unique stationary distribution from primitives. The model is deliberately one-sided: exit occurs through exogenous attrition at rate δ, so the equilibrium is a theory of activation timing and replacement pressure rather than of endogenous exit. The empirical exercises use the net establishment entry rate (entry minus exit) as the model-consistent counterpart of $\dot{n}$; combining timing search

with Jovanovic (1982)-style learning is a natural extension that would endogenize exit (Section 6).

Remark 1 (Isochrony)**.**
The frequency ω does not depend on the amplitude A. Small and large business cycles have the same frequency. The economic reason is that both the speed of adjustment and the distance to travel scale with the size of the displacement. A large shock creates strong congestion pressure, driving rapid correction—but the economy must also traverse a longer path (overshooting proportionally far past n_p before returning). A small shock creates weaker pressure and slower adjustment, but the path is shorter. Since speed and distance both scale linearly with A, the time per cycle $T = 2\pi\sqrt{\mu/\varphi}$ is independent of severity. Isochrony extends to the damped equilibrium: successive swings preserve their period even as the envelope decays (Figure 1).

Structural decomposition The frequency $\omega = \sqrt{\varphi/\mu}$ decomposes into two primitives. The numerator φ is the curvature of the profit landscape: how sharply profits fall when the industry deviates from optimal scale. The denominator μ is the timing friction: how costly it is to adjust the number of active firms rapidly, arising from congestion in the entry matching process. Treated as an abstract pair, (φ, μ) alone determines ω; the frequency does not depend on productivity z (which affects profit levels but not curvature), on the realized exit rate δ, or on the shock process (Section 4). When μ is calibrated structurally via Proposition 1, $\mu = (1-\alpha)/[\alpha(\delta n_p)^2]$ inherits dependence on the matching elasticity α and the steady-state throughput δn_p; the resource-stock R underlying the matching technology is absorbed into the steady-state calibration of n_p and does not appear separately in the formula. The structural decomposition is therefore: ω depends on (φ, μ) as primitives, and on (α, δ, n_p) when μ is read off the microfoundation. Table 1 contrasts this decomposition with the way standard frameworks pin cycle frequency.

3.5 Properties of the cycle

Corollary 1 (Properties)**.** *Under Theorem 1: (i) $\partial\omega/\partial\varphi > 0$ and $\partial\omega/\partial\mu < 0$; (ii) ω is independent of productivity z, the shock process σ_ε, and the amplitude A; (iii) the period is $T = 2\pi\sqrt{\mu/\varphi}$; (iv) (*frequency-welfare separation*) $\omega_0 = \sqrt{\varphi/\mu}$ depends only on market-structure primitives (φ, μ), while the welfare cost (Section 6.2) depends only*

Table 1: What determines cycle frequency?

	Frequency determined by	**Cross-industry prediction**
RBC	Shock autocorrelation	Common if shocks are common
NK	Policy rule	Common across industries
Beaudry et al. (2020)	Bifurcation condition	Not addressed
Timing search	$\boldsymbol{\omega = \sqrt{\varphi/\mu}}$	**Industry-specific frequency**

Notes: In RBC and NK models, the spectral peak reflects the shock process or the policy rule—objects common across industries. In Beaudry et al. (2020), the frequency emerges from a Hopf bifurcation that depends on the full parameterization. In timing search, the frequency is a closed-form function of two market-structure primitives (φ: profit-landscape curvature; μ: timing friction), is exact under linear utility, and is isolated from the shock process (Proposition 6).

on the orthogonal pair $(\sigma_\varepsilon, \chi)$ — frequency and welfare cost are governed by disjoint primitives, with cycle severity $\sigma_x = \sigma_\varepsilon/\sqrt{2\chi\varphi}$ as the intermediate object that depends on $(\sigma_\varepsilon, \chi, \varphi)$; (v) the amplitude A is free: the system is a center, and the stationary distribution is selected by the stochastic extension (Section 4).

The orbit decomposes into four phases: scale-externality-driven boom, overshoot above n_p from matching-friction inertia, congestion-driven bust, and undershoot below n_p. All three ingredients — scale externality, congestion, and timing friction — are needed; removing any one eliminates the oscillation. The ironing condition (13) is structurally identical to energy conservation, with profit-landscape loss as potential energy and adjustment cost as kinetic energy, which is why it generates closed orbits rather than monotone convergence.

3.6 Beyond the Gaussian: general profits

The Gaussian profit function delivers exact results, but the mechanism does not depend on it. This subsection shows that the frequency formula extends to general hump-shaped profits, quantifies the approximation error, and derives a new prediction about cycle asymmetry.

Proposition 3 (General hump-shaped profits)**.**
Under any C^2 profit function $\pi(n)$ with $\pi'(n_p) = 0$ and $\pi''(n_p) < 0$, and a timing

friction $\xi(\dot{n}) = e^{-\frac{\mu}{2}\dot{n}^2}$*, the ironing condition*

$$\ln \pi(n) - \frac{\mu}{2}\dot{n}^2 = E$$

admits periodic orbits in the potential $U(n) = -\frac{1}{\mu}\ln \pi(n)$*, which captures how the profit landscape confines the economy near optimal scale. The frequency near* n_p *is:*

$$\omega = \sqrt{\frac{|f'(n_p)|}{\mu}} + O(A^2), \tag{16}$$

where $f'(n_p) = \pi''(n_p)/\pi(n_p) < 0$*.*

Proof. See Online Appendix B.2. ■

For the Gaussian, $|f'(n_p)| = \varphi$ exactly. For $\pi(n) = n^\alpha(a - bn)$, $|f'(n_p)| = \alpha/n_p^2 + b^2/(a - bn_p)^2$. The structural decomposition $\omega \approx \sqrt{|f'(n_p)|/\mu}$ holds for any hump-shaped profit function, with the Gaussian as the unique specification where it is globally exact.

A natural concern is that the exactness is a knife-edge property. The following result characterizes the correction.

Proposition 4 (Frequency correction under asymmetric profits)**.**
For a C^4 *profit function with* $\pi'(n_p) = 0$ *and* $\pi''(n_p) < 0$*, the oscillation frequency at amplitude* A *is:*

$$\omega(A) = \omega_0\big(1 + c_4\, A^2 + O(A^4)\big), \tag{17}$$

where $\omega_0 = \sqrt{|f'(n_p)|/\mu}$ *and* c_4 *depends on* $U^{(3)}(n_p)$ *and* $U^{(4)}(n_p)$*, with* $U(n) = -\frac{1}{\mu}\ln \pi(n)$ *the effective potential. For the Gaussian,* $c_4 = 0$ *exactly (isochrony).*

Proof. See Online Appendix B.4. ■

A numerical check across 8 smooth exponential-family hump-shaped profit specifications shows that the Gaussian-based prediction $T_0 = 2\pi\sqrt{\mu/\varphi}$ stays within 1.6% of the exact orbit-integrated period at $A/n_p = 0.30$ (Online Appendix C.2). The Gaussian is therefore not the necessary specification but the analytically tractable benchmark; the closed-form frequency is robust to general hump-shaped profits at empirically relevant amplitudes.

The Gaussian also produces exactly symmetric cycles, which is counterfactual—empirical business cycles feature slower expansions and sharper contractions. Relaxing symmetry turns this weakness into a prediction.

Proposition 5 (Cycle asymmetry)**.**
When $\pi'''(n_p) \neq 0$, the periodic orbit is asymmetric. The leading-order asymmetry parameter—the fractional difference in time spent above versus below n_p—is:

$$\mathcal{A} \equiv \frac{t_{above} - t_{below}}{t_{below}} = \frac{\pi'''(n_p)}{4\,|\pi''(n_p)|}\,A + O(A^2). \tag{18}$$

When $\pi'''(n_p) < 0$ (profit falls faster on the congestion side), the economy spends more time below n_p: slow expansions and sharp contractions.

Proof. See Online Appendix B.4. ■

The degree of asymmetry is pinned by the ratio $\pi'''(n_p)/|\pi''(n_p)|$, which is independently measurable from the curvature and skewness of industry profit functions. Online Appendix C.2 illustrates the asymmetric orbit numerically and verifies the anharmonic frequency correction (Proposition 4) against simulations under quartic and skewed profit perturbations.

4 Stochastic Extension

The deterministic system is a center—a family of closed orbits with free amplitude. Small stochastic perturbations reveal the resonance of this center; the damped extension (Section 4.3, Proposition 7) then selects a unique stationary distribution by combining shocks with Rayleigh damping.

4.1 Adding shocks

Consider an i.i.d. disturbance to profits:

$$\tilde{\pi}(n,t) = \pi(n) + \sigma_\varepsilon\,\varepsilon_t, \qquad \varepsilon_t \overset{\text{i.i.d.}}{\sim} (0,1). \tag{19}$$

The ironing condition no longer holds exactly: E drifts stochastically according to $\mathrm{d}E \approx \frac{\sigma_\varepsilon\,\varepsilon_t}{\pi(n)}\dot{n}\,\mathrm{d}t$, and the economy migrates between orbits.

4.2 Properties

Proposition 6 (Stochastic limit cycle)**.**
Under small shocks (19)*:*

(i) The spectral density of $n(t)$ peaks at the structural frequency $\omega_0 = \sqrt{\varphi/\mu}$ in the undamped small-shock limit; with Rayleigh damping $\eta = \chi/\mu > 0$, the observed PSD peak is $\omega_{PSD} = \sqrt{\varphi/\mu - \eta^2/2}$, equal to ω_0 up to an $O(\eta^2)$ correction (Proposition 7(iv)).

(ii) The peak frequency is determined by (φ, μ) at leading order and by the damping ratio η at second order, not by σ_ε.

(iii) Even i.i.d. shocks produce persistent cycles: the persistence is endogenous to the equilibrium structure.

Proof sketch. Write the stochastic system in action–angle coordinates (A, ϕ), where $n = n_p + A\cos\phi$ and $\dot{n} = -A\omega_0 \sin\phi$. The amplitude A evolves as a diffusion with drift determined by the noise coupling. The spectral density of $n(t)$ is the power spectrum of a stochastically forced harmonic oscillator; its transfer function has a resonance peak at the structural frequency $\omega_0 = \sqrt{\varphi/\mu}$ in the undamped limit and at $\omega_{\text{PSD}} = \sqrt{\varphi/\mu - \eta^2/2}$ under Rayleigh damping, independent of σ_ε in either case. The full derivation is in Online Appendix B.1. ■

The structural frequency is a property of the market structure, not of the shock process: i.i.d. shocks produce an observed PSD peak at $\omega_{\text{PSD}} = \sqrt{\omega_0^2 - \eta^2/2}$, while persistent shocks shift the observed peak slightly below this (negligible whenever the transfer function is sharp relative to the shock persistence). This matches the central finding of Beaudry et al. (2020)—persistent boom-bust dynamics from short-lived disturbances—though the mechanism here is the scale-congestion trade-off rather than financial-friction complementarities. A numerical Euler-Maruyama simulation of the stochastic system confirms the predicted spectral concentration; Online Appendix B.1 and C.1 report the full derivation, the shock-spectrum analysis, and the phase-portrait diagnostics.

4.3 The operative equilibrium: damped stochastic oscillator

In the operative equilibrium—with Rayleigh bottleneck dissipation and profit shocks—the industry behaves as a damped, stochastically forced oscillator at the same natural frequency $\omega = \sqrt{\varphi/\mu}$. Dissipation from convex congestion in the entry-replacement pipeline pulls the industry toward optimal scale, shocks re-excite it, and the stationary distribution of $(n - n_p)$ has variance $\sigma_\varepsilon^2/(2\eta\varphi\mu)$ (Proposition 7). Theorem 1's closed ellipse is the frictionless limit ($\eta \to 0$, $\sigma_\varepsilon \to 0$); the operative equilibrium is the economically relevant extension.

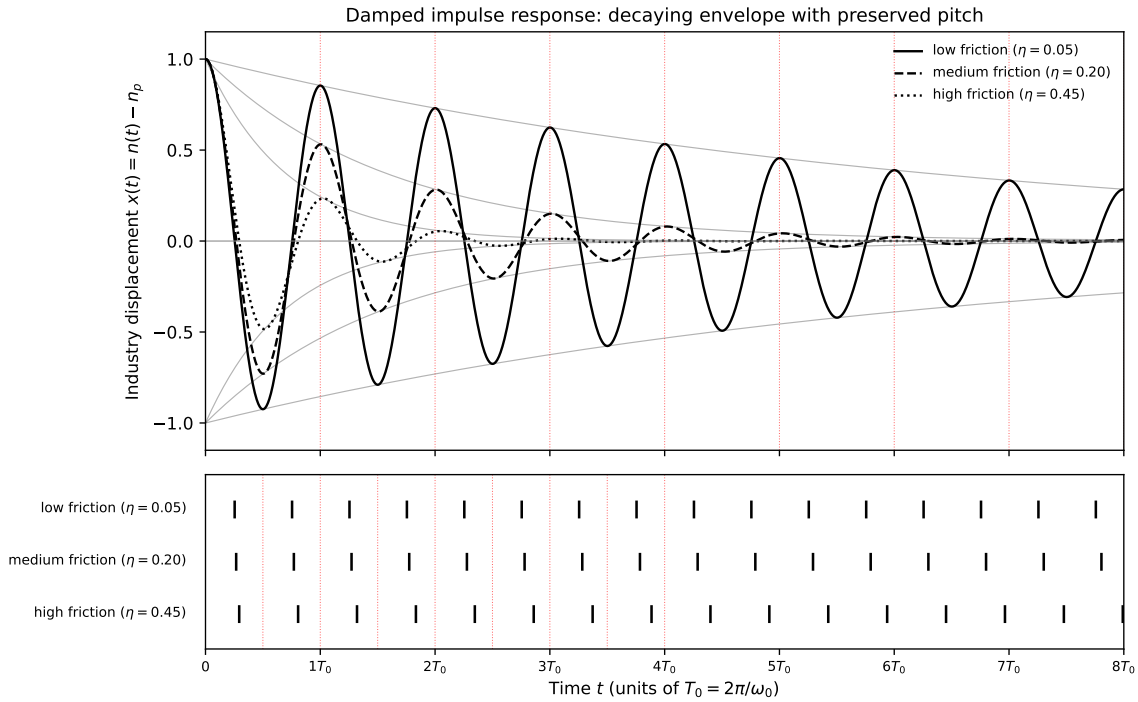
The entry-replacement bottleneck consumes real resources—permits, specialized equipment, site development, reorganization services, write-downs—at a rate that is convex in $\dot{n}$ around steady state (Online Appendix B.5). The convex part of this cost, with Rayleigh friction coefficient χ at steady-state flow, dissipates output at a rate $\chi\dot{n}^2$, while shocks with standard deviation σ_ε continuously perturb the industry. Together, these two forces turn the closed orbit of Theorem 1 into the damped stochastic oscillator (12). Online Appendix B.5 derives the damped equation in two steps. First, differentiating the ironing condition gives the conservative Newton equation $\mu\ddot{n} + \varphi(n - n_p) = 0$. Second, balancing the rate at which conservative energy is lost against the bottleneck's output-dissipation rate $\chi\dot{n}^2$ yields the additional drag term $\chi\dot{n}$, with signs fixed by the requirement that the bottleneck consumes rather than produces output:

$$\ddot{n} + \eta\,\dot{n} + \frac{\varphi}{\mu}(n - n_p) = \frac{\sigma_\varepsilon}{\mu}\,\varepsilon_t, \tag{20}$$

with $\eta = \chi/\mu$ emerging endogenously from the Rayleigh friction coefficient χ and the matching friction μ of Proposition 1. The shock σ_ε in (20) is the reduced-form intensity of the linearized acceleration shock obtained by differentiating the ironing condition around $(n_p, 0)$; the state-dependence of the original profit-shock-to-energy-drift map vanishes at this linearization (Online Appendix B.5). The natural frequency $\omega = \sqrt{\varphi/\mu}$ is unchanged: damping and shocks shape when and how severely the industry oscillates, not its structural pitch.

Figure 1 illustrates the impulse-response behaviour of (20) in the deterministic-but-damped case ($\sigma_\varepsilon = 0$, $\eta > 0$, $x(0) = A_0 \neq 0$), which is the case that the empirical echo analysis in Section 5.4 examines in manufacturing investment. Three different damping rates η are plotted on the same axes, with fixed (φ, μ) so that the natural frequency is common across the three trajectories. Each trajectory's envelope decays

Figure 1: Damped impulse response of the timing-search equilibrium



Notes: The damped harmonic oscillator $\ddot{x} + \eta\,\dot{x} + (\varphi/\mu)\,x = 0$ with $x(0) = A_0$, $\dot{x}(0) = 0$, and fixed natural frequency $\omega_0 = \sqrt{\varphi/\mu} = 1$. Three damping rates $\eta \in \{0.05, 0.20, 0.45\}$ are plotted. Top panel: trajectories with the exponential envelope $A_0 e^{-\eta t/2}$ shown in grey; the dotted red grid marks integer multiples of $T_0 = 2\pi/\omega_0$. Bottom panel: zero crossings of each trajectory, showing that the half-period marker (red) coincides with each crossing. Higher friction damps the envelope faster but leaves the pitch ω_0 essentially unchanged. The empirically relevant signature—decaying amplitude with preserved period—is documented in the manufacturing investment echo of Section 5.4.

exponentially at rate $\eta/2$, but the zero crossings occur at nearly identical times: the pitch ω is preserved while the volume fades. This decoupling of declining amplitude from preserved period is a distinctive feature of the timing-search equilibrium and contrasts with linear models such as AR(1) in which the decay rate and the period are mechanically linked.

Proposition 7 (Amplitude selection)**.**
With Rayleigh friction coefficient $\chi > 0$ (damping $\eta = \chi/\mu > 0$) and shocks $\sigma_\varepsilon > 0$:

(i) Without shocks ($\sigma_\varepsilon = 0$): orbits spiral inward to $(n_p, 0)$ with decay half-life $\ln 2 \cdot 2\mu/\chi$.

(ii) With shocks: the system has a unique stationary distribution with variance:

$$\mathbb{E}\big[(n-n_p)^2\big] = \frac{\sigma_\varepsilon^2}{2\,\eta\,\varphi\,\mu} = \frac{\sigma_\varepsilon^2}{2\,\chi\,\varphi}, \tag{21}$$

where the second equality uses $\eta = \chi/\mu$, so that the matching friction μ cancels.

(iii) The Hilbert envelope $A = \sqrt{(n-n_p)^2 + (\dot{n}/\omega_0)^2}$ follows a Rayleigh distribution with scale $\sigma_x = \sigma_\varepsilon/\sqrt{2\chi\varphi}$, so that $\mathbb{E}[A^2] = 2\sigma_x^2 = \sigma_\varepsilon^2/(\chi\varphi)$.

(iv) The spectral peak frequency of the driven oscillator's stationary spectrum is $\omega_{peak} = \sqrt{\varphi/\mu - \eta^2/2}$, obtained by maximizing the transfer function $|H(\omega)|^2 = 1/[(\omega_0^2-\omega^2)^2 + \eta^2\omega^2]$ over ω. (The homogeneous damped natural frequency $\sqrt{\varphi/\mu - \eta^2/4}$, at which the unforced solution oscillates, is a separate object.) For small η, both quantities are negligibly below $\sqrt{\varphi/\mu}$.

Proof. See Online Appendix B.5. ■

Substituting $\eta = \chi/\mu$ gives $\sigma_x = \sigma_\varepsilon/\sqrt{2\chi\varphi}$, in which μ cancels exactly: industries with volatile profits, low Rayleigh friction, or flat profit landscapes experience larger cycles, while μ governs only the cycle frequency. Combining with the welfare formula (25):

$$\mathbb{E}[\mathcal{L}] \approx \frac{\varphi}{2}\,\mathbb{E}[A^2] = \frac{\sigma_\varepsilon^2}{2\,\chi}. \tag{22}$$

Higher Rayleigh friction χ *reduces* welfare loss by damping the cycle—a "stabilization through friction" result—with the welfare cost depending only on $(\sigma_\varepsilon, \chi)$ as φ cancels. Online Appendix C.1 reports the phase-portrait diagnostics and the numerical verification of the stationary Rayleigh distribution.

4.4 Frequency-welfare separation

The closed-form structural frequency $\omega_0 = \sqrt{\varphi/\mu}$ (Theorem 1), the stationary variance $\sigma_x^2 = \sigma_\varepsilon^2/(2\chi\varphi)$ (Proposition 7 after substituting $\eta = \chi/\mu$, so that μ cancels), and the welfare cost $\mathbb{E}[\mathcal{L}] = \sigma_\varepsilon^2/(2\chi)$ (Section 6.2) together establish the paper's central structural claim: *structural frequency and welfare cost have disjoint determinants*, with cycle severity standing as an intermediate observable depending on the overlapping set $(\sigma_\varepsilon, \chi, \varphi)$. We state this as a single theorem.

Theorem 2 (Frequency-welfare separation).
In the timing search equilibrium under Assumptions 1–3 and the damped-stochastic specification of Proposition 7, with $\eta = \chi/\mu$:

$$\begin{aligned} \textit{structural frequency:} \quad & \omega_0 = \sqrt{\varphi/\mu} && \textit{depends only on } (\varphi, \mu); \\ \textit{cycle severity:} \quad & \sigma_x = \sigma_\varepsilon/\sqrt{2\,\chi\,\varphi} && \textit{depends on } (\sigma_\varepsilon, \chi, \varphi); \\ \textit{welfare cost:} \quad & \mathbb{E}[\mathcal{L}] = \sigma_\varepsilon^2/(2\chi) && \textit{depends only on } (\sigma_\varepsilon, \chi). \end{aligned}$$

The structural frequency depends on (φ, μ); the welfare cost depends on $(\sigma_\varepsilon, \chi)$; these determinant sets are disjoint, with cycle severity the intermediate observable connecting both. The structural frequency is invariant to σ_ε, χ, the shock persistence (in the i.i.d. specification), and the realized cycle amplitude; the welfare cost is invariant to (φ, μ) and to the per-firm entry cost κ. The observed PSD peak inherits the structural frequency up to an $O(\eta^2)$ correction: $\omega_{PSD} = \sqrt{\varphi/\mu - \eta^2/2}$ (Proposition 7(iv)).

Proof. The structural-frequency claim is Theorem 1 (deterministic linearized dynamics admit eigenfrequency $\omega_0 = \sqrt{\varphi/\mu}$, depending on (φ, μ) alone). The severity claim follows by substituting $\eta = \chi/\mu$ into Proposition 7(ii): $\sigma_x^2 = \sigma_\varepsilon^2/(2\eta\varphi\mu) = \sigma_\varepsilon^2/(2\chi\varphi)$, so μ cancels exactly. The welfare claim follows from $\mathcal{L} \approx (\varphi/2)A^2$ in (25) and the Rayleigh-envelope identity $\mathbb{E}[A^2] = 2\mathbb{E}[(n - n_p)^2] = \sigma_\varepsilon^2/(\chi\varphi)$; substituting gives $\mathbb{E}[\mathcal{L}] = (\varphi/2)\mathbb{E}[A^2] = \sigma_\varepsilon^2/(2\chi)$, with the dependence on φ canceling. The observed-PSD-peak claim is Proposition 7(iv). ■

The theorem has three immediate methodological consequences. First, cross-industry variation in ω_0 identifies the market-structure ratio φ/μ, with no contamination from differences in shock processes, the Rayleigh friction coefficient χ, or the per-firm entry cost κ (the $O(\eta^2)$ shift to the observed PSD peak is small at the empirical damping levels of Section 5, $\eta^2/(2\omega_0^2) \lesssim 5\%$); the individual primitives φ and μ require external information, as discussed in Section 5.3. The cross-sectional regression of Section 5.3 therefore tests a structural prediction about φ/μ even though the proxies for the individual primitives are imperfect. Second, two industries with identical market structure (φ, μ) but different shock environments cycle at the same frequency in the model. This makes the model's cross-industry frequency prediction structural rather than a reduced-form artifact of common shocks propagating through industry-specific filters. Third, industrial policy that targets μ (entry regulation) or

φ (cluster development) affects timing alone in the model; only policies that move σ_ε or χ (the Rayleigh friction coefficient) affect welfare. The policy implications in Section 6.3 follow directly from this separation.

Scope of the separation: leakage under alternative welfare metrics Theorem 2's welfare claim relies on three conditions: (i) linear utility, (ii) the per-firm effective-output welfare metric in equation (25), and (iii) excluding the supplier's $\frac{1}{2}\chi\dot{n}^2$ resource flow from the welfare metric (Section 2.5). Relaxing each of these reintroduces some leakage of (φ, μ) into the welfare cost. We document the leakage numerically in Online Appendix E.1 and summarize here.

Under CRRA utility with risk-aversion coefficient $\sigma_{\text{ra}} \in \{1, 2, 5\}$, the consumption-equivalent welfare loss departs from the linear-utility baseline $\sigma_\varepsilon^2/(2\chi)$ by +8% to +11% at the median across a representative grid of $(\varphi, \mu, \chi, \sigma_\varepsilon)$, with maximum leakage of about ±100% in the worst cells (high φ or low χ, where amplitude is large and risk aversion bites the consumption density most). Under an alternative welfare metric that adds the supplier's resource flow to the per-firm output drop, the welfare cost becomes $\mathbb{E}[\mathcal{L}^{\text{alt}}] = \sigma_\varepsilon^2/(2\chi) + \sigma_\varepsilon^2/(2\mu)$, which explicitly reintroduces μ. The median leakage is +20% across the grid, reaching +80% at low μ. We therefore state the separation as exact under conditions (i)–(iii), with median leakage of order 10–25% under standard CRRA preferences or under welfare metrics that include the supplier's resource flow. The separation is exact, not approximate, in the welfare framework adopted here, but is not robust to the specific welfare metric chosen, and we make this caveat explicit.

Comparison to alternative endogenous-cycle models Timing search and Beaudry et al. (2020) share the abstract architecture of a stochastic limit cycle but differ in mechanism (BGP's financial frictions and Hopf bifurcation versus timing search's agglomeration-congestion trade-off and Rayleigh bottleneck dissipation) and, methodologically, in tractability: the closed-form $\omega_0 = \sqrt{\varphi/\mu}$ is industry-decomposable, which is what makes the cross-industry regression of Section 5.3 feasible. A side-by-side simulation against AR(1) RBC, New Keynesian AR(2), and Beaudry et al. (2020) (Van der Pol limit cycle) benchmarks confirms that only timing search jointly delivers cross-industry frequency heterogeneity, post-shock periodicity at industry-specific frequencies, and trend overshoot (Online Appendix F.1). A more demanding multi-

sector RBC with input-output linkages (Acemoglu et al., 2012) and industry-specific AR(2) productivity shocks can reproduce the observed cross-industry dispersion $\sigma(\ln \omega_i) = 0.36$ but cannot reproduce the empirical slope: across six IO/AR specifications the simulated $\hat{\beta}_\gamma$ centers near zero with 95% CIs roughly $[-0.13, +0.20]$, and zero of 50 replications in any cell reach the empirical slope estimates in the $\hat{\beta}_\gamma \in [0.41, 0.48]$ range across the multitaper, Burg AR(12), and IV specifications (Online Appendix F.2). The granular-shock channel of Gabaix (2011) provides a related alternative absorbed into the multi-sector specification. The same logic rules out Khan and Thomas (2008)-style industry-heterogeneous capital-adjustment costs: γ does not enter that model either, so the simulated slope is structurally near zero (Online Appendix F.4). The discriminating signature against both alternatives is the structural slope β_γ, not the cross-industry dispersion. Online Appendix F.3 provides the side-by-side mechanism table.

5 Quantitative Framework

Because the model predicts characteristic frequencies rather than deterministic cycles, the empirical analysis asks whether industry fluctuations contain systematic frequency differences associated with the model's primitives. Theorem 1's closed form $\omega = \sqrt{\varphi/\mu}$ implies a log-linear cross-industry regression with structural slope 1/2 on $\ln \varphi$; under matching friction approximately uniform within manufacturing, this is what the data should reveal in the cross-section.

The model delivers three testable cross-industry predictions. First, cycle frequency should vary with market-structure proxies, not with the shock process. Second, large one-time shocks should trigger deterministic oscillations at each industry's structural frequency, not at a common frequency inherited from the shock. Third, industries displaced by a common shock should overshoot their pre-shock trend rather than converge monotonically. Section 5.3 tests the first prediction using the Ellison-Glaeser concentration index across 16 manufacturing 3-digit industries, with a parallel cross-section across 10 BDS subsectors using independently measured spatial concentration as a robustness extension. Section 5.4 tests the second and third predictions through three diagnostics: post-shock periodicity in manufacturing and oil investment using Fisher's g-test, BDS net-entry oscillation features across 19 sectors, and the post-COVID overshoot.

5.1 Cross-industry predictions

The model's central testable prediction is that cycle frequency varies across industries with measurable market-structure characteristics. Taking logs of $\omega_i = \sqrt{\varphi_i/\mu_i}$ yields a regression framework:

$$\ln \omega_i = \frac{1}{2} \ln \varphi_i - \frac{1}{2} \ln \mu_i + u_i, \tag{23}$$

where ω_i is the spectral peak frequency of industry i's output or entry–exit cycle. The coefficients $\frac{1}{2}$ on $\ln \varphi$ and $-\frac{1}{2}$ on $\ln \mu$ are *exact* predictions of the model—not free parameters—so equation (23) is testable with industry-level proxies for scale-externality sharpness and adjustment friction.

The qualitative predictions are as follows:

(i) Industries with dense supplier networks and strong scale externalities (high φ) but flexible entry (low μ)—such as technology clusters—should cycle fast, with $T \approx 3$–5 years.

(ii) Manufacturing industries with moderate linkages and moderate frictions should cycle at intermediate frequencies, with $T \approx 5$–10 years.

(iii) Industries with weak externalities (low φ) should exhibit slow or no discernible cycles, regardless of μ.

(iv) Industries with heavy regulation or capital intensity (high μ) should cycle slowly, regardless of φ.

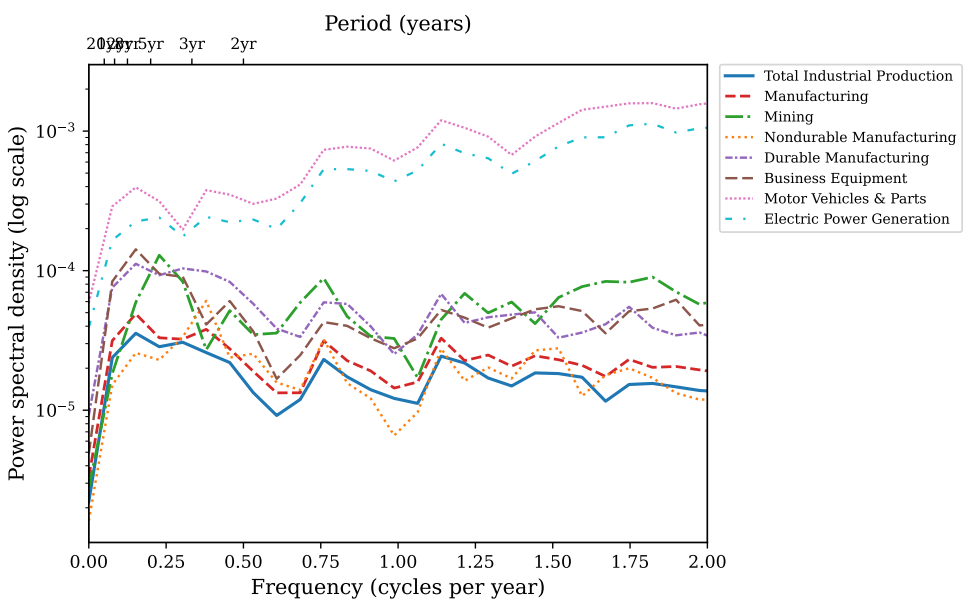
Crucially, two industries with identical shock processes but different (φ, μ) will cycle at different frequencies. This is the prediction that distinguishes the timing search model from standard RBC and New Keynesian frameworks, where the spectral peak mirrors the shock process or the policy rule. The headline cross-industry test of Section 5.3 narrows to 16 manufacturing 3-digit industries, within which the matching technology (permits, specialist construction labor, equipment supply chains) is broadly common and turnover varies only 1.7×; under approximate μ uniformity in this sample, predictions (i)–(iv) reduce to a single φ-driven slope, $\beta_\gamma = 1/2$.

5.2 Spectral evidence from U.S. industry data

I evaluate the cross-industry prediction using monthly industrial production indices from FRED for eight U.S. sectors over 1972–2024. The series span a range of market structures: from Motor Vehicles & Parts (dense supplier networks, strong scale externalities) to Mining (dispersed extraction, weak inter-firm linkages) to Electric Power Generation (regulated, capital-intensive).

Figure 2 plots the power spectral density of log-differenced production for each industry. Two features stand out. First, spectral peaks vary substantially across industries. Motor Vehicles, Durable Manufacturing, Business Equipment, and aggregate Manufacturing concentrate spectral mass near periods of 6–7 years. Mining peaks at 4.4 years. Nondurable Manufacturing and Electric Power peak near 2.6 years. Second, the cross-industry clustering is informative: industries sharing supply-chain linkages—durables, motor vehicles, business equipment—cluster at similar frequencies despite producing very different goods. This is consistent with ω depending on market structure, not on product-specific shocks.

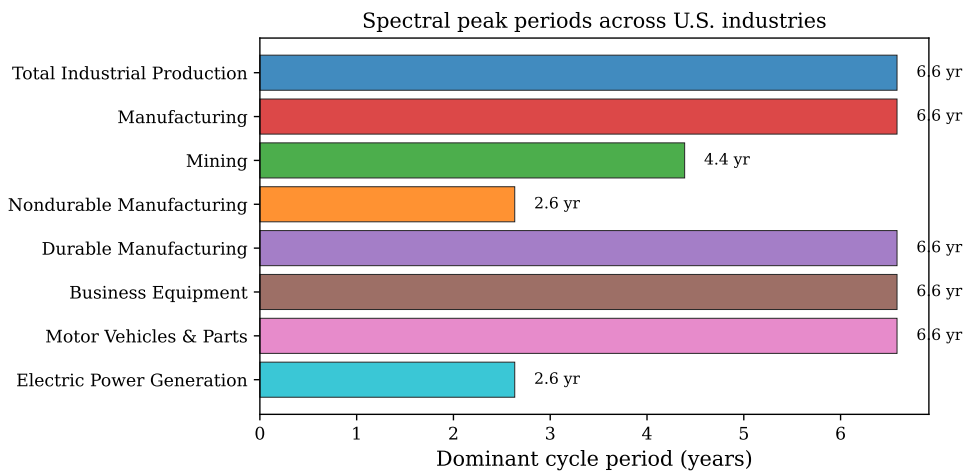
Figure 2: Spectral density of U.S. industry production growth



Notes: Power spectral density (Welch's method) of monthly log-differenced industrial production indices from FRED, 1972:M1–2024:M12. The top axis labels convert frequency (cycles per year) to period (years). Industries with strong supply-chain linkages (Motor Vehicles, Business Equipment, Durable Manufacturing) cluster near $T \approx 6$–7 years, while industries with different market structures (Mining, Nondurables, Electric Power) show distinct peak frequencies.

Figure 3 summarizes the dominant cycle period for each industry, defined as the reciprocal of the spectral peak frequency in the business-cycle band (periods between 1.5 and 30 years). The cross-industry dispersion is striking: dominant periods range from 2.6 years (Electric Power, Nondurable Manufacturing) to 6.6 years (Durable Manufacturing, Motor Vehicles). This two-and-a-half-fold variation in cycle frequency across industries that share a common macroeconomic environment—the same monetary policy, the same aggregate demand shocks, the same financial conditions—is difficult to reconcile with models that derive spectral properties from a common shock process. In the timing search framework, the variation is structural: it reflects cross-industry differences in market-structure primitives. Across this broad eight-industry sample—which spans Manufacturing, Mining, and Electric Power—both the profit-landscape curvature (φ_i) and the timing friction (μ_i) plausibly vary across sectors with different matching technologies. The headline test of Section 5.3 narrows to 16 manufacturing 3-digit industries, within which μ is approximately uniform, so the cross-industry signal collapses to the φ channel.

Figure 3: Cross-industry variation in spectral peak periods

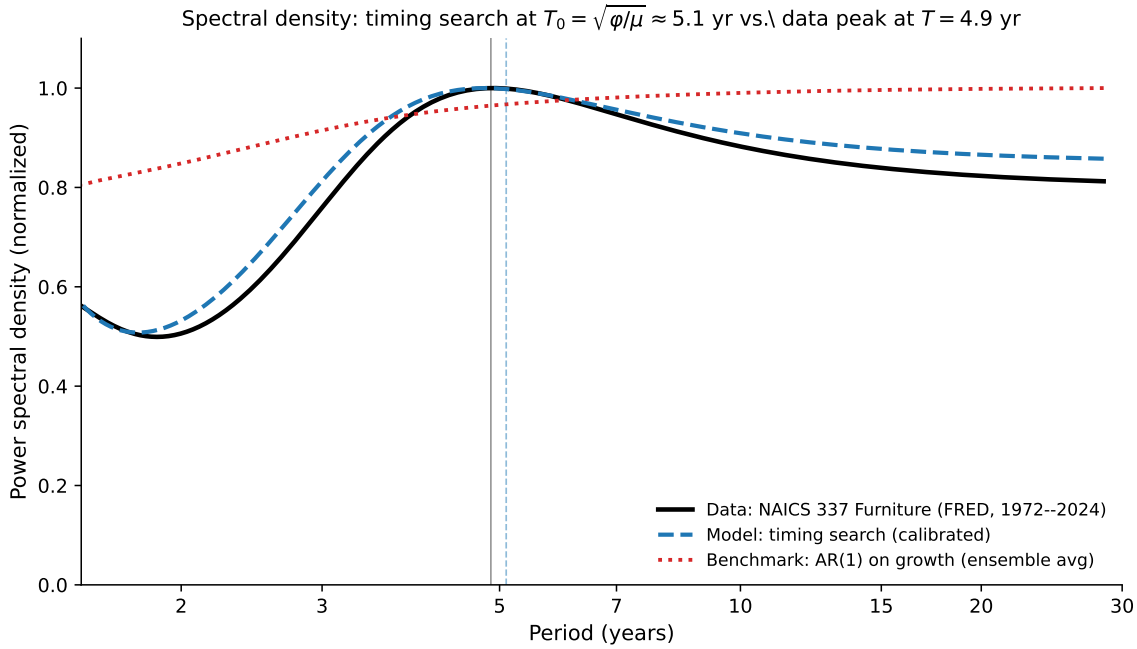


Notes: Dominant cycle period (years) for each U.S. industry, defined as the reciprocal of the spectral peak frequency in the business-cycle band (periods between 1.5 and 30 years). Industries with strong scale–congestion externalities (Motor Vehicles, Durable Manufacturing, Business Equipment) cluster near $T \approx 6$–7 years. Industries with weaker or different externality structures (Mining, Nondurables, Electric Power) cycle at distinct frequencies.

Model shape versus data shape The timing-search SDE generates an internal spectral peak at $\omega_0 = \sqrt{\varphi/\mu}$, while an AR(1) on growth rates produces a smooth

low-pass spectrum with no internal peak. Figure 4 compares the model PSD at a manufacturing-like calibration ($T_0 = 5.0$ years) to the FRED industrial production PSD for NAICS 337 (Furniture), which peaks at $T = 4.9$ years: the model can produce a shape comparable to the data at a single industry, providing qualitative motivation for the cross-industry slope test below.

Figure 4: Spectral density: timing-search model vs. industry data



Notes: Burg AR(24) power spectral densities, normalized to the maximum within the business-cycle band (1.5–30 years). Solid black: FRED monthly industrial production growth for NAICS 337 (Furniture), 1972–2024, with empirical peak at $T = 4.9$ years. Dashed blue: simulated timing-search stochastic differential equation at a manufacturing-like calibration ($\varphi = 1.52$, $\mu = 1$, $\kappa = 0.32$, $\sigma_\varepsilon = 0.04$, theoretical $T_0 = 2\pi/\sqrt{\varphi/\mu} = 5.1$ years; empirical Burg-AR peak at $T = 4.8$ years), ensemble-averaged over 30 replications, with realistic-conditions noise components (idiosyncratic measurement noise $\sigma_{\text{meas}} = 0.060$ and a slow AR(1) productivity-drift component, $\rho = 0.93$, $\sigma = 0.0011$) that mimic the finite-sample data generating process. Dotted red: ensemble average of 30 AR(1)-on-growth-rate replications ($\rho = 0.5$), shown for comparison—a standard "no internal peak" benchmark with smooth low-pass spectrum. Timing search can generate a comparable peak and surrounding spectral shape; AR(1) cannot.

Consistency check: nonlinear dependence in industry dynamics As a consistency check rather than a discriminating test, we apply the BDS test (Brock et al., 1996) to AR(1)-filtered industry residuals; the i.i.d. null is rejected at the 1% level for all eight industries with z-statistics ranging from 16.2 (Durable Manufacturing)

to 21.8 (Motor Vehicles & Parts). This establishes that industry dynamics contain nonlinear or nonstationary structure beyond AR(1), which is consistent with the timing-search mechanism but is also consistent with GARCH, regime switching, and structural breaks (the 1972–2024 sample spans the Great Moderation and COVID, both of which can produce comparable BDS rejection magnitudes). The BDS exercise therefore confirms departure from linear AR(1) dynamics but does not discriminate among the nonlinear alternatives. Full results across embedding dimensions 2–6, with AR(1) and AR(p)+GARCH pre-filtering specifications, are in Online Appendix D.2.

5.3 Cross-industry regression with external proxies

A baseline regression on seven broad industry groups using BEA proxies (inverse I-O self-purchase ratio for φ and capital per employee for μ) gives a univariate slope of 0.85 ($R^2 = 0.65$, $p = 0.028$) on $\ln \hat{\varphi}_i$ and a multivariate R^2 of 0.75, motivating the broad-sector relationship (Online Appendix D.1). The headline test scales this up: $N = 16$ three-digit manufacturing industries with the Ellison and Glaeser (1997) geographic concentration index γ_i as the φ proxy and Burg AR(24) spectral peaks as ω_i.[2]

Table 2 reports the results. In the univariate specification, the Ellison and Glaeser (1997) concentration index significantly predicts cycle frequency ($\hat{\beta}_\gamma = 0.20$, $p = 0.010$, $R^2 = 0.39$, $N = 16$). The sign matches the theory: geographically concentrated industries—those with stronger agglomeration externalities (higher φ)—cycle faster.

Two refinements close most of the apparent gap to the theoretical slope $\beta_\gamma = 0.5$. First, alternative spectral estimators yield slope estimates closer to theory: a multitaper PSD (Thomson DPSS, $K = 5$ tapers) gives $\hat{\beta}_\gamma = 0.41$ ($p = 0.004$, $R^2 = 0.47$), and Burg AR(12) gives $\hat{\beta}_\gamma = 0.48$ ($p = 0.029$). The Burg AR(24) baseline with $\hat{\beta}_\gamma = 0.20$ is the most conservative point estimate; multitaper and shorter-order Burg both deliver coefficients within one standard error of the theoretical 0.5 (Online Appendix D.6 reports the full six-estimator panel). Second, instrumenting the modern γ_i with historical concentration patterns (Hoover localization coefficients in 1900, 1927, and 1947 from Kim (1995)) yields a 2SLS estimate of $\hat{\beta}_\gamma^{\text{IV}} = 0.45$ (HC1 SE 0.16). Standard Wald inference rejects $\beta_\gamma = 0$ at $p = 0.015$ and fails to reject the theoretical $\beta_\gamma = 0.5$

[2]The proxy swap is empirical: at the 3-digit level, the broad-sector I-O and capital-intensity proxies do not preserve the relationship; the EG index, constructed at the 3-digit manufacturing level by design, does. Online Appendix D.1 documents.

at $p = 0.77$; the Hansen overidentification test passes ($p = 0.96$). The first-stage $F = 1.55$ is below the conventional Stock-Yogo weak-IV threshold, so we also report Anderson-Rubin weak-IV-robust inference: AR-based testing strongly fails to reject the structural $\beta_\gamma = 0.5$ ($p = 0.99$) and rejects $\beta_\gamma = 0$ only marginally ($p = 0.06$). Pre-1947 industrial geography pre-dates the modern macroeconomic environment in which cycle frequencies are observed, motivating the exclusion restriction; we acknowledge the standard caveat that historical concentration could correlate with persistent industry traits that affect modern cycle frequency through channels other than current γ. We read the IV evidence as corroborative of the structural slope rather than as independently identifying: the AR confidence set is wide given the weak first stage, and the IV result is best interpreted alongside the OLS, multitaper, and BDS subsector estimates as a coherent body of evidence pointing to β_γ in the neighborhood of the theoretical 1/2. The recovered slope has a sharper structural reading: the model predicts $\omega_i = \sqrt{\varphi_i/\mu_i}$, so the regression slope on $\ln \gamma_i$ is 1/2 only if $\ln \mu_i$ is approximately uncorrelated with $\ln \gamma_i$ across the cross-section. Within manufacturing, this is plausible on first principles: the matching technology (permits, specialist construction labor, equipment supply chains) is a generic property of the entry-matching pipeline rather than an industry-specific friction, and the structural formula $\mu_i = (1-\alpha)/[\alpha(\delta_i n_{p,i})^2]$ produces narrow within-manufacturing variation when α is approximately uniform and turnover δ_i ranges only 1.7× ($\delta \in [0.065, 0.11]$). The empirical recovery of the theoretical slope $\beta_\gamma \approx 1/2$ is therefore consistent with μ approximately uniform across manufacturing industries: cross-industry frequency variation operates through the profit-landscape curvature φ, which has an order-of-magnitude wider range (γ varies from 0.004 to 0.12, a 30× range). External μ proxies that mix friction with industry size, asset durability, or regulatory burden are wrong-signed (Online Appendix D.5), consistent with the interpretation that the *friction-relevant* component of μ is not what those proxies measure. The economic content of approximate μ-uniformity within manufacturing is that input factors relevant to the entry-matching technology—permits, specialist construction labor, equipment supply chains, contractor networks—flow flexibly across 3-digit manufacturing industries; what differentiates them is not their access to entry resources but the agglomeration density and supplier-network structure they enter into.

Three robustness exercises further support the cross-sectional relationship without overturning it. First, a parallel $N = 10$ cross-section across BDS subsectors with clear

Table 2: Structural regression: geographic concentration and cycle frequency

	OLS $N = 16$	OLS multitaper $N = 16$	2SLS (historical IV) $N = 15$	Theory
$\hat{\beta}_\gamma$	0.20	0.41	0.45	0.50
HC1 SE	(0.07)	(0.12)	(0.16)	—
Wald p vs. 0	0.010	0.004	0.015	—
Wald p vs. 0.5	< 0.001	0.46	0.77	—
AR p vs. 0	—	—	0.06	—
AR p vs. 0.5	—	—	0.99	—
R^2 / 1st-stage F	0.39	0.47	$F = 1.55$	—

Notes: Cross-industry regression of $\ln \omega_i$ on $\ln \gamma_i$ (Ellison and Glaeser 1997 geographic concentration index). Column 1: OLS with Burg AR(24) spectral peaks (paper baseline). Column 2: OLS with multitaper spectral peaks (Thomson DPSS, $K = 5$, Thomson 1982). Column 3: 2SLS with historical Hoover localization coefficients (1900, 1927, 1947 from Kim 1995) as instruments for $\ln \gamma_i$; Hansen overidentification $p = 0.96$. Spectral peaks estimated from log-differenced monthly industrial production (FRED, 1972–2024). HC1 heteroskedasticity-robust standard errors (White, 1980). AR rows report Anderson-Rubin weak-IV-robust p-values, which apply to column 3 only. The first-stage $F = 1.55$ is below the conventional Stock-Yogo weak-IV threshold; column 3 should be read as corroborative of the structural slope rather than independently identifying.

interior spectral peaks, using BDS-derived spatial concentration G_i rather than the published γ_i, yields $\hat{\beta} = 0.107$ (HC1 SE 0.043, $p = 0.037$, $R^2 = 0.39$): same sign, similar R^2, different data source for both ω_i and the concentration measure (Online Appendix D.7; sample selection on a feature of the dependent variable, exploratory only). Second, a pooled US+UK cross-section using UK ONS Index of Production data (1990–2024) for 12 NACE-2 manufacturing divisions matched to the US 16 yields a pooled slope $\hat{\beta}_\gamma = 0.13$ ($p = 0.026$) with country fixed effects; the UK-only slope is small and insignificant, reflecting the strong assumption that US γ values approximate UK industry-specific concentration (UK BRES-derived γ would address this but requires non-trivial data construction; Online Appendix D.7). Third, at four-digit resolution ($N = 86$), Ellison et al. (2010) publish γ only at the three-digit level, so every four-digit child inherits its parent's γ and the slope mechanically collapses to $\hat{\beta}_\gamma = 0.027$ ($p = 0.705$); this tests the inheritance of γ rather than its predictive power at four-digit resolution (Online Appendix D.4).

Identification The ratio φ/μ is identified directly from the spectral peak ($\varphi/\mu \approx (2\pi/T^*)^2$ at leading order, with an $O(\eta^2)$ correction). Individual φ and μ require

external information; χ is identified from $\sigma_x = \sigma_\varepsilon/\sqrt{2\chi\varphi}$ given $(\sigma_\varepsilon, \varphi)$; and κ enters only the steady-state ironing level.

5.4 Additional diagnostics

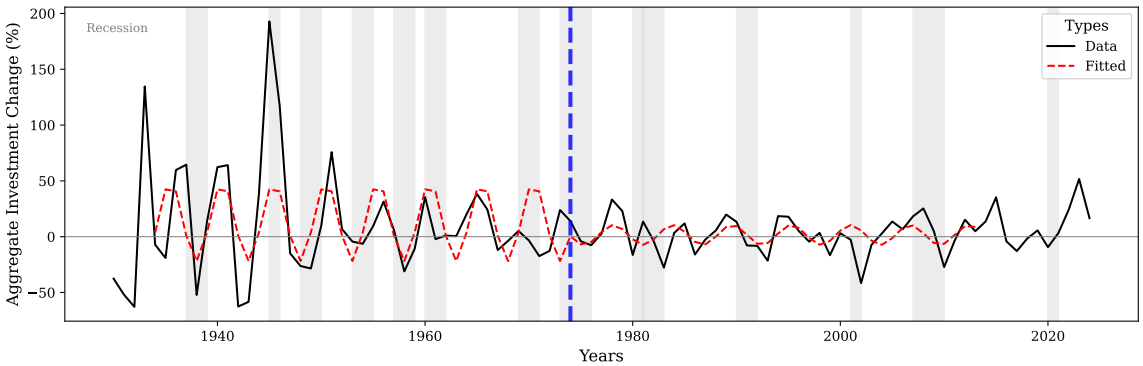
The cross-sectional regression in Section 5.3 is the paper's structural test, identifying the φ channel through the geographic-concentration proxy. We complement it with three additional diagnostics that are consistent with the timing-search mechanism but rest on weaker identification: oscillatory features in BDS net entry rates that lie outside the AR(1) class, post-shock periodicity in two historical investment series at industry-specific structural frequencies, and an aggregate post-COVID overshoot pattern. We present these as suggestive rather than definitive evidence; their joint pattern—interior spectral peaks across sectors, post-shock periodicity at industry-specific frequencies, and trend overshoot after a common displacement—is harder to reproduce in any single alternative model than in any one diagnostic taken alone.

Oscillatory entry dynamics across industries (BDS, 1978–2022) For each of 19 NAICS sectors, we compute the linearly detrended net establishment entry rate (entry rate minus exit rate) and examine two AR(1)-distinguishing features. First, four sectors (Construction, Wholesale, Retail, Real Estate) exhibit significantly negative autocorrelation at lag 4 outside the 95% Bartlett band, a pattern incompatible with any scalar AR(1) process (which requires non-negative ACF at even lags). Second, Fisher's g-test (Fisher, 1929) on the detrended series rejects the null of no deterministic periodicity at the 5% level in 10 of the 19 sectors, with spectral peaks concentrated near $T \approx 8$–9 years. The sign-changing autocorrelation pattern is also harder to reconcile with vintage-capital replacement, which produces positive entry waves at vintage age rather than alternation between net entry above trend and net exit above trend. Online Appendix D.2 provides the full methodology, sector-by-sector results, and figure.

Post-shock periodicity in historical impulse responses A second diagnostic uses BEA investment-in-structures data (1930–2024) to test whether large one-time shocks trigger oscillations at industry-specific structural frequencies. Manufacturing investment after the Great Depression/WWII oscillates at $T = 5.0$ years with striking visual regularity (Figure 5; Fisher's g-test $p = 0.036$ in the 1934–1973 echo window;

$p = 0.578$ in the 1974–2013 non-echo window), while mining/oil investment after the 1979 crisis oscillates at $T = 3.25$ years (Figure 6; $p = 0.016$). The detected period is invariant to ±3-year window shifts; only marginal p-values vary at the boundaries. The cross-industry frequency difference—manufacturing ringing slowly through 40 years, oil ringing rapidly through 25—is consistent with different (φ, μ) across industries and harder to reconcile with explanations that would produce a common frequency. The vintage-capital alternative for manufacturing (Kydland and Prescott, 1982b; Cooper et al., 1999) is compatible with rather than competing with the timing-search interpretation: replacing wartime capital is itself a wave of establishment-level construction subject to the same timing frictions. Online Appendix D.3 provides the full data construction, methodology, robustness across windows, and figures.

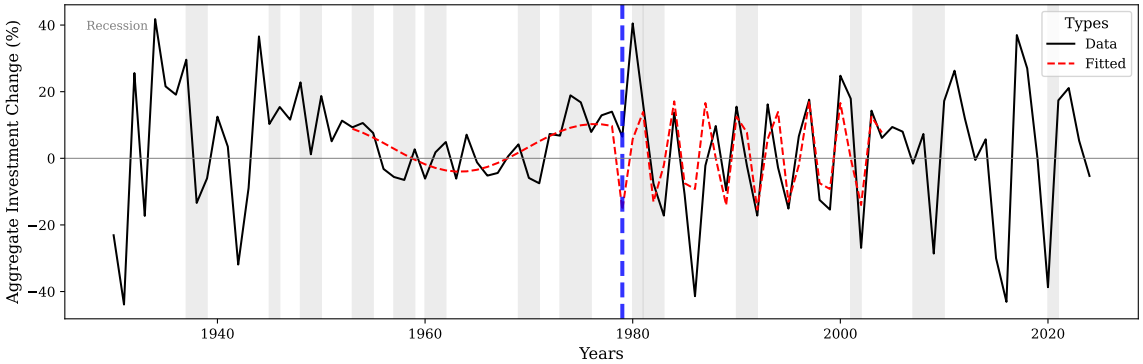
Figure 5: Post-shock periodicity: manufacturing investment



Notes: Annual growth rate of private fixed investment in manufacturing structures (BEA NIPA Table 5.4.1, line 16). Thick solid line: data. Red dashed line: fitted cosine from Fisher's g-test. Gray bands: NBER recessions. Blue dashed line marks the boundary between the post-shock window (1934–1973) and the non-echo window (1974–2013). The fitted cosine tracks the data closely in the post-shock window ($T = 5.0$ years, $p = 0.036$), capturing the regular boom-bust pattern following the Great Depression and WWII mobilization.

The post-COVID natural experiment The COVID-19 pandemic displaced all U.S. industries below trend in 2020 simultaneously. Timing search predicts that displaced industries should not return monotonically to trend but rather overshoot the pre-shock level: the ironing condition forces the economy past optimal scale before cycling back. In the BDS data, 14 of 19 NAICS sectors (74%) overshoot above their pre-COVID linear trend in 2021 or 2022, against an AR(1)-monotone-convergence baseline of approximately 30% calibrated to sector-specific pre-COVID

Figure 6: Post-shock periodicity: oil and mining investment



Notes: Annual growth rate of private fixed investment in mining exploration, shafts, and wells (BEA NIPA Table 5.4.1, line 24). Blue dashed line marks the 1979 oil crisis. Post-crisis oscillations at $T \approx 3.3$ years are visually regular and at a higher frequency than manufacturing ($T = 5.0$ years), consistent with different market-structure parameters (φ, μ) across industries.

residual variance; a binomial test against the 30% null rejects at $p < 0.001$. The test is qualitative: it asks whether industries return to trend monotonically (AR(1)) or overshoot (timing search), without invoking cross-sectional structural variation in the response magnitude.[3] The post-COVID entry boom—widely discussed in policy circles as an anomalous recovery—is exactly the overshooting phase that the timing-search cycle predicts.

6 Discussion

The closed-form frequency $\omega = \sqrt{\varphi/\mu}$ and its separation from the welfare cost of cycles rest on five maintained assumptions: a hump-shaped profit landscape, the Gaussian specification, linear utility, a convex matching friction, and exogenous exit. This section addresses each in turn, showing that the cycle mechanism survives relaxations while the closed-form exactness is Gaussian-specific. It then computes the welfare cost of the cycles and the implications for industrial policy, where the frequency-welfare separation delivers concrete predictions for CHIPS-Act-style entry subsidies.

[3]The structural prediction that overshoot magnitude correlates with sector-level concentration is not borne out cross-sectionally in the BDS sample for the COVID shock. The variation in observed overshoot is dominated by sector-specific COVID exposure (lockdown intensity, demand shifts, supply-chain disruption), violating the model's "common shock" idealization at the sector level even though the binary aggregate prediction holds clearly. A clean cross-sectional natural-experiment test of the structural channel would require a shock more uniform across sectors than COVID was.

6.1 Modeling choices and limitations

Four limitations bound the scope of the closed-form result. (i) *Gaussian exactness*: the symmetric-cycle prediction of the Gaussian specification is counterfactual; under general hump-shaped profits (Proposition 3), cycles become asymmetric and the frequency formula is accurate to 1.6% at empirically relevant amplitudes (Proposition 4; Online Appendix C.2). (ii) *Exogenous exit*: the model treats exit through the constant-rate δ rather than as a learning- or productivity-driven endogenous decision; combining timing search with Jovanovic (1982)-style learning is a natural extension. (iii) *Linear utility and the per-firm welfare metric*: under CRRA preferences or under welfare metrics that include the supplier's resource flow, (φ, μ) leak into the welfare cost with median magnitude 10–25% (Online Appendix E.1), so the frequency-welfare separation is exact under our maintained metric and approximately exact otherwise. (iv) *Within-manufacturing scope of the empirical identification*: the headline test isolates the φ channel under the maintained identifying restriction that μ is approximately uniform across manufacturing 3-digit industries. This is a maintained restriction rather than a directly identified property; three suggestive lines of evidence are consistent with it, none decisive on its own. First, external μ proxies—capital intensity, asset durability, regulatory burden—deliver wrong-signed coefficients in the cross-industry regression (Online Appendix D.5), which is consistent with μ-uniformity but could equally reflect that those proxies fail to capture the friction-relevant component of μ. Second, the recovered $\hat{\beta}_\gamma \approx 0.45$ is broadly consistent with the structural 1/2 that would obtain under μ-uniformity (Wald $p = 0.77$, Anderson-Rubin $p = 0.99$); given the weak first-stage ($F = 1.55$), we read this as corroborative rather than identifying. Third, the structural formula $\mu_i = (1 - \alpha)/[\alpha(\delta_i n_{p,i})^2]$ predicts narrow within-manufacturing variation given approximate α-homogeneity in matching technology and a 1.7× turnover range, providing theoretical consistency with the maintained restriction. Cross-industry tests outside manufacturing (where μ may genuinely vary, as in the broader $N = 10$ subsector cross-section) require an industry-friction-specific proxy at NAICS-3 resolution—permit-to-occupancy lead times by industry × state, specialist construction-labor tightness—that current public data do not provide; this direct test of the μ channel is left for future data work. Equivalently, input factors relevant to entry matching are largely fungible across 3-digit manufacturing industries; what varies cross-industrially is the profit landscape firms enter, not the matching technology that delivers them. Online Appendix A.3 derives

μ from discrete-time proportional adjustment costs as an alternative microfoundation.

6.2 Welfare cost of endogenous cycles

The ironing condition provides a natural welfare measure. Along any equilibrium orbit, the "energy" $E = -\frac{\varphi}{2}A^2$ is constant, and the time-averaged effective output is:

$$\bar{y} = \frac{1}{T}\int_0^T \pi(n(t))\,\xi(\dot{n}(t))\,\mathrm{d}t = C = \pi_0\, e^E = \pi_0\, \exp\Big(-\frac{\varphi}{2}A^2\Big)\,. \tag{24}$$

Since $A > 0$ in any stochastic equilibrium, $\bar{y} < \pi_0 = \pi(n_p)\xi(0)$: the time-averaged per-firm effective output is strictly below the optimal-scale, zero-adjustment level. The proportional per-firm effective-output welfare loss from cycling is:

$$\mathcal{L} = 1 - \frac{\bar{y}}{\pi_0} = 1 - e^{-\varphi A^2/2} \approx \frac{\varphi}{2}\,A^2 \quad \text{for small } A. \tag{25}$$

The welfare cost depends on two structural parameters: φ (the curvature of the profit landscape, determining how costly deviations from optimal scale are) and A (the cycle amplitude, selected in stochastic equilibrium by the shock variance σ_ε^2, the Rayleigh friction χ, and the profit-landscape curvature φ). Crucially, $\mathcal{L}$ does not depend on μ. The timing friction determines the *frequency* of cycles but not their welfare cost. Two industries with the same φ and A but different μ cycle at different frequencies yet suffer identical welfare losses.

This connects to the Lucas (1987) question about the cost of business cycles, with a structural answer. In the Lucas framework, the welfare cost depends on the variance of consumption, taken as a reduced-form object. Here, the welfare cost is a function of measurable primitives: $\mathbb{E}[\mathcal{L}] \approx (\varphi/2)\,\mathbb{E}[A^2] = \varphi\,\mathbb{E}[(n-n_p)^2]$. The curvature φ can be estimated from the profit landscape, and the envelope A (or equivalently $\mathbb{E}[(n-n_p)^2]$) from observed entry–exit cycles. The cost is zero only if $\sigma_\varepsilon = 0$ (no shocks) or $\chi = \infty$ (perfect damping). In the stochastic equilibrium, $\mathbb{E}[(n-n_p)^2] = \sigma_\varepsilon^2/(2\chi\varphi)$, so the expected welfare loss scales as $\sigma_\varepsilon^2/\chi$—independent of the cycle frequency.

Welfare metric and the role of χ The welfare measure is the time-averaged drop in per-firm effective output $\pi(n)\xi(\dot{n})$ relative to its steady-state value at $(n, \dot{n}) = (n_p, 0)$, as in equation (25): $\mathcal{L} \approx (\varphi/2)A^2$ in the small-amplitude limit. The Rayleigh friction χ enters this measure only through the equilibrium amplitude: in the linear damped

oscillator with white-noise forcing (Proposition 7), the stationary envelope-squared is $\mathbb{E}[A^2] = \sigma_\varepsilon^2/(\chi\varphi)$ (equivalently, $\mathbb{E}[(n-n_p)^2] = \sigma_\varepsilon^2/(2\chi\varphi)$, with $\mathbb{E}[A^2] = 2\mathbb{E}[(n-n_p)^2]$ for the Rayleigh envelope), so the proportional welfare loss reduces to $(\varphi/2)\mathbb{E}[A^2] = \sigma_\varepsilon^2/(2\chi)$ in the small-amplitude limit. The supplier-side resource flow $\frac{1}{2}\chi\dot{n}^2$ derived in Section 2.5 is a separate accounting object, real but not part of this welfare metric; its time-average (which depends on μ) describes the resources consumed by the entry-matching infrastructure rather than the deviation of effective output from optimal scale. Tracking the resource flow as a separate object preserves the closed form $\mathbb{E}[\mathcal{L}] = \sigma_\varepsilon^2/(2\chi)$ for the per-firm welfare loss while leaving the supplier-cost accounting available for policy exercises that wish to net it out.

Numerical welfare cost per industry Calibrating χ_i for each manufacturing industry from $\sigma_{x,i} = \sigma_{\varepsilon,i}/\sqrt{2\chi_i\varphi_i}$ and computing $\mathbb{E}[\mathcal{L}_i] = \sigma_{\varepsilon,i}^2/(2\chi_i)$ as a fraction of $\pi_{0,i}$ yields a mean welfare cost of 0.39% of π_0 across 10 cached manufacturing industries, with range $[0.12\%, 0.70\%]$—comparable to Lucas (1987)- and Reis (2009)-style consumption-equivalent business-cycle welfare costs (typically 0.1 to 1%). Online Appendix E.2 reports the per-industry table.

6.3 Industrial policy and the anatomy of recovery

Two policy implications follow from the frequency-welfare separation (Theorem 2), both under the per-firm effective-output welfare metric. First, the structural parameters act on identifiable margins: μ governs timing alone, φ governs timing and severity, χ governs severity and per-firm welfare, σ_ε governs severity and per-firm welfare, and κ governs only the steady-state ironing level. In the stochastic equilibrium, the per-firm welfare cost reduces to $\mathbb{E}[\mathcal{L}] \approx (\varphi/2)\mathbb{E}[A^2] = \sigma_\varepsilon^2/(2\chi)$, with both φ and μ canceling and κ absent from the formula. Second, under this metric the Rayleigh friction $\chi > 0$ is *welfare-improving*: $\chi \to 0$ drives $\mathcal{L} \to \infty$, while $\chi > 0$ pins finite severity. Selective deregulation that targets μ alone delivers faster cycles without the amplitude penalty; blanket deregulation that also reduces χ amplifies cycles. The qualitative "stabilization through χ" claim is robust to alternative welfare metrics that include the supplier resource flow (Online Appendix E.1); only the clean μ-independence of $\mathbb{E}[\mathcal{L}]$ depends on excluding it.

The model contains two structurally distinct entry-cost objects: κ, the per-firm sunk cost paid at entry, and χ, the Rayleigh friction summarizing bottleneck-strain

dissipation. The welfare formula depends on χ alone; κ enters the ironing level but not the welfare loss. A CHIPS-Act-style subsidy that halves χ for NAICS 3344 (calibrated baseline period $T = 4.75$ years) leaves cycle frequency unchanged, raises σ_x by $\sqrt{2}$, and doubles the per-firm welfare cost; a cluster-development policy that raises φ by 20% shortens T by 9% and lowers σ_x by the same factor with welfare unchanged; streamlined permitting that reduces μ by 30% shortens T by 16% with σ_x and welfare unchanged; a κ-only transfer moves none of these. Mapping a real-world entry-cost intervention onto χ rather than κ requires an institutional argument that the policy changes the bottleneck-strain mechanism. Online Appendix E.3 reports the full counterfactual table and the policy taxonomy.

The qualitative claim that some real-world "entry friction" is stabilizing is conditional on the friction loading onto χ rather than onto κ.

6.4 Extensions

Several extensions merit investigation. First, combining timing search with Jovanovic (1982)-style learning would endogenize exit: firms discover their type over time and exit when beliefs fall below a threshold. Such a synthesis would produce entry waves followed by shakeouts, with the entry–exit lag governed by learning speed, generating rich cross-sectional dynamics jointly with the aggregate cycle. Second, the linear-utility assumption makes industries independent; under non-linear preferences (e.g., log utility), industries interact through the marginal utility of aggregate consumption, coupling their dynamics. With finitely many industries or correlated shocks, this coupling could produce richer aggregate fluctuations, though the closed-form frequency formula would likely not survive. Third, the parameters φ and μ are taken as exogenous primitives. Endogenizing them through industry investment in linkages (affecting φ) or entry infrastructure (affecting μ) would generate a richer set of predictions. Fourth, the interaction between the timing search mechanism and financial frictions (Beaudry et al., 2020) could produce models with multiple spectral peaks: one from market structure and one from financial amplification. Finally, the framework extends naturally to any setting where agents choose *when* to act and face a trade-off between positive and negative externalities. Applications include hiring waves, investment cycles, technology adoption, and financial market dynamics.

7 Concluding Remarks

This paper provides a structural account of cross-industry frequency heterogeneity. Under *timing search*—agents choosing *when* to enter a market subject to matching frictions—the equilibrium ironing condition forces overshoot rather than monotone adjustment, and the closed-form cycle frequency $\omega_0 = \sqrt{\varphi/\mu}$ emerges in two market-structure primitives: profit-landscape curvature φ and entry-matching friction μ. The headline empirical test brings this prediction to U.S. industry data: across 16 manufacturing 3-digit industries, the cross-industry slope $\hat{\beta}_\gamma$ falls in the $[0.41, 0.48]$ range across estimators (multitaper, Burg AR(12), and IV with historical concentration from Kim (1995) instrumenting for current γ), broadly consistent with the structural prediction $\beta_\gamma = 1/2$. Given a weak first stage ($F = 1.55$), we read the IV as corroborative rather than identifying; under Anderson-Rubin weak-IV-robust inference, $1/2$ is not rejected ($p = 0.99$). Multi-sector RBC (Acemoglu et al., 2012) and Khan and Thomas (2008)-style heterogeneous capital-adjustment costs reproduce cross-industry frequency dispersion but not the empirical slope across estimators. Cycle length, in this account, is a property of how firms interact within an industry, not of the shocks hitting it.

A second result is a *frequency-welfare separation*: $\omega_0 = \sqrt{\varphi/\mu}$ depends only on the market-structure primitives $\{\varphi, \mu\}$, while the per-firm effective-output welfare cost $\mathcal{L} = \sigma_\varepsilon^2/(2\chi)$ depends only on the disjoint pair $\{\sigma_\varepsilon, \chi\}$. The separation emerges from the same ironing condition rather than being imposed, and it delivers an instrument-to-outcome map for industrial policy: μ governs timing alone, φ governs both timing and severity, χ governs severity and per-firm welfare, and κ governs only the ironing level (Section 6.3).

The within-manufacturing identification rests on a maintained identifying restriction—μ approximately uniform across 3-digit manufacturing industries—with three suggestive lines of support. External μ proxies (capital intensity, asset durability, regulatory burden) deliver wrong-signed coefficients in the cross-industry regression (Online Appendix D.5), which is consistent with μ-uniformity but could equally reflect those proxies failing to capture the friction-relevant component of μ. The recovered slope's broad consistency with $1/2$ corroborates this reading under weak-IV inference. The structural μ formula and the narrow turnover range supply theoretical consistency. The economic interpretation is that entry inputs—permits, construction labor, equip-

ment supply chains—flow flexibly across manufacturing industries, while the profit landscape firms enter does not; what drives the cross-industry signal is the latter. A direct cross-industry test of the μ channel using industry-friction-specific proxies remains for future data work. Beyond entry timing, the framework extends to any setting in which agents choose *when* to act under matching frictions—capital investment, hiring, technology adoption, financial-market participation—each with its own market-structure-pinned frequency.

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